



Dr.-Ing. KO WATANABE

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Trustworthy AI · HCI Researcher

« Postdoctoral Researcher at DFKI, specializing in trustworthy AI, HCI, multimodal sensing, medical AI, and learning analytics. »

Skills

Research

- Eye Tracking
- Deep learning
- Multimodal sensing
- Trustworthy AI · Medical AI
- HCI study design

Engineering

- Python · JavaScript / TypeScript
- Swift · Kotlin · Ruby on Rails
- AWS / GCP · Docker
- Tobii · Pupil Labs

Languages

Japanese (native) · English (business level)
· German (daily use)

Key highlights

KI4Pol — SDS Group

Conducted R&D in the KI4Pol joint research project with the National Police Agency (details undisclosed due to NDA).

Mission KI — SDS Group

Lead developer of evaluation tooling for medical AI products in the federal Mission KI initiative.

LeCycl Project — SDS Group

Worked in education x AI on a learning support system designed for effective knowledge transfer.

ACM ETRA 2025 (BEST SHORT PAPER AWARD)

Received the Best Short Paper Award at ACM ETRA 2025 for research on webcam based pupil diameter estimation.

Interests

Communities

Peer review and organizing: CHI, ETRA, AHCs, ISWC/UbiComp, IUI, MUM, ABC, ICMU, SmartComp, CVPR, ML4H

Application areas

Medical AI assurance, learning analytics, and multimodal interfaces for trustworthy HCI

Professional experience

Postdoctoral Researcher — DFKI GmbH	Dec 2024 – Present
Chief Technology Officer (CTO) — LOGOT UG	May 2024 – Present
Affiliate Assistant Professor (Concurrent) — NAIST	Apr 2026 – Present
Visiting Researcher (Concurrent) — OMU	May 2022 – Present
Technical Advisor — Mesh Inc.	Dec 2024 – Aug 2025
Technical Advisor — Affectify	Mar 2024 – Jun 2025
Software Engineer (MLOps) — Bcode	Sep 2024 – Dec 2024
Software Engineer (Backend Engineer) — Ichiroku	Feb 2024 – Dec 2024
Doctoral Researcher — DFKI / RPTU PsyberLab	Mar 2021 – Nov 2024
Software Engineer — DeNA Co., Ltd. (Pococha)	Apr 2019 – Mar 2021
Development Instructor (Swift / iOS) — TechPlayAcademy	Nov 2020 – Dec 2020
Software Engineer — Rakuten Group Inc.	Jan 2018 – Mar 2018

Education

RPTU Kaiserslautern-Landau — Germany	2021 – 2024
Ph.D. in Computer Science — Advisors: Prof. Andreas Dengel, Associate Prof. Shoya Ishimaru	
Nara Institute of Science and Technology (NAIST) — Japan	2017 – 2019
M.S. in Computer Science — Advisors: Prof. Keiichi Yasumoto, Associate Prof. Yutaka Arakawa	
Tokyo University of Agriculture and Technology — Japan	2013 – 2017
B.S. in Mechanical Engineering — Advisors: Prof. Venture Gentine	

Selected publications

- Towards Facilitated Fairness Assessment of AI-based Skin Lesion Classifiers Through GenAI-based Image Synthesis — WACV 2026
- PupilSense: A Novel Application for Webcam-Based Pupil Diameter Estimation — ETRA 2025 (BEST SHORT PAPER AWARD)
- EyeUnderstand: Dashboard for Gaze and Deep-Learning Driven Comprehension Estimation in Online Lectures — IEEE Access, 2025
- TKG-DM: Training-free Chroma Key Content Generation Diffusion Model — CVPR 2024 (HIGHLIGHT PAPER)
- Engauge: Engagement Gauge of Meeting Participants Estimated by Facial Expression and Deep Neural Network — IEEE Access, 2023

Honors

- Awards: Best Short Paper Award, ACM ETRA 2025
- Funding: CREST AIP Challenge · TOBITATE Exchange Scholarship · JASSO Scholarship